DEEP RESEARCH REPORT

AI-Driven Web Accessibility & WCAG Compliance

How Artificial Intelligence can be incorporated and utilized to make web accessibility and WCAG compliance more efficient, accurate, faster, and user-friendly — a foundational brief and use-case catalogue for an AI/ML engineering team.

Prepared for: The AI/ML Engineering Team, Edvensoft
Audience: AI/ML engineers exploring accessibility use cases
Date: August 2026
Scope: Foundations • Legal & business case • Application areas • Non-AI best practice • 7 AI use cases with deeply-profiled market leaders

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1. How to read this report & a note on sources

This report answers the six questions your team set, in order. The first four build shared context (what accessibility is, why it exists, where it applies, and how the industry does it today without AI). The fifth and sixth — the heart of the brief — catalogue seven high-impact places where AI is already being used, each written in your requested presentation format and each followed by a deep profile of the two-to-three market leaders so you can review them yourself.

> **Everything here is written in plain language but stays technically precise.** Where a technical term matters (POUR, ARIA, ASR, VLM), it is defined in-line the first time it appears. Nothing is dumbed down; it is just explained.

A note on how this research was done and how trustworthy it is

Facts were gathered by searching the open web with a priority order of **official sources first** (W3C/WAI, government bodies such as the U.S. DOJ, the EU, WHO), **then** primary vendor documentation and product pages, **then** reputable industry articles and blogs (Deque, WebAIM, Level Access, UsableNet, and similar). Version numbers, dates, and legal facts were checked against primary sources. Every vendor link was resolved during research. A small number of claims that could not be fully verified are flagged in-line with the word *flag* or a caution box, so you always know the difference between a confirmed fact and a reasonable-but-unconfirmed one. The full source list is in the final section.

> **One honest theme runs through the whole report:** across almost every AI use case, the credible pattern is **“AI for scale + a human expert for correctness.”** AI dramatically speeds up and broadens accessibility work, but current technology does not deliver guaranteed WCAG conformance on its own. The report is candid about where AI genuinely solves the problem, where it only assists, and where over-claiming has already led to regulatory penalties.

2. Q1 — What is Web Accessibility and WCAG Compliance?

Web accessibility, in one idea

Web accessibility means designing and building websites, web apps, mobile apps, and digital documents so that **people with disabilities can perceive, understand, navigate, interact with, and contribute** to them on an equal footing with everyone else. In practice it also means the content works correctly with **assistive technologies** — screen readers (which read the screen aloud), screen magnifiers, braille displays, voice-control software, switch devices, and alternative keyboards.

Who actually benefits (it is not a niche)

Accessibility spans a wide range of needs, permanent and temporary:

- **Visual** — blindness, low vision, colour blindness. These users rely on screen readers, magnification, high contrast, and text alternatives for images.
- **Auditory** — deaf and hard-of-hearing users, who rely on captions for video and transcripts for audio.
- **Motor / physical** — limited fine motor control, tremors, paralysis, or repetitive strain injury. These users often cannot use a mouse and rely on the keyboard, large targets, or voice control.
- **Cognitive & learning** — dyslexia, ADHD, autism, memory or attention differences. These users benefit from plain language, consistent layouts, and reduced distraction.
- **Speech** — affects people who use voice interfaces.
- **Situational & temporary** — a broken arm, glare on a phone in sunlight, a noisy train (needing captions), a slow connection. These affect **everyone at some point**, which is why accessibility helps the general population too.

What WCAG is

WCAG stands for the **Web Content Accessibility Guidelines**. It is published by the **W3C** (the World Wide Web Consortium) through its **Web Accessibility Initiative (WAI)**, and it is the **international technical standard** for what makes web content accessible. Almost every accessibility law and procurement rule in the world points to WCAG. It is deliberately *technology-agnostic*: it describes *outcomes* ("images need text alternatives") rather than dictating specific code.

How WCAG is structured: the POUR principles

WCAG is a hierarchy: **4 principles → guidelines → success criteria**. The four principles are remembered by the acronym **POUR**:

Principle	What it means	Examples
Perceivable	Users must be able to sense the information with whatever senses they have.	Alt text for images, captions for video, sufficient colour contrast.
Operable	Users must be able to operate the interface.	Everything works by keyboard, no keyboard traps, enough time, no seizure-triggering flashes.

Understandable	Content and operation must be clear and predictable.	Readable text, consistent navigation, helpful error messages.
Robust	Content must work reliably across browsers, devices, and assistive tech, now and in future.	Valid, well-structured code; correct use of ARIA.

Under these principles sit **guidelines** (13 of them in WCAG 2.2), and under each guideline sit **success criteria** — the specific, *testable* pass/fail statements that make conformance measurable.

Conformance levels: A, AA, AAA

Each success criterion is assigned a level:

- **Level A** — the most basic, essential requirements. Failing these creates major barriers.
- **Level AA** — the practical target for almost everyone, and the level that nearly all laws reference. When something is called "compliant," it almost always means **WCAG 2.1 AA** or **WCAG 2.2 AA**.
- **Level AAA** — the highest bar. The W3C explicitly says it is **not recommended as a blanket requirement** for whole sites, because not all content can satisfy every AAA criterion.

Levels are cumulative: meeting AA means you also meet A.

The versions, and where each stands in 2026

Version	Published	Status as of August 2026
WCAG 2.0	Dec 11, 2008	Still valid (also an ISO standard, ISO/IEC 40500), but superseded in practice.
WCAG 2.1	Jun 5, 2018	Still valid and still the version most laws cite. Added 17 criteria focused on mobile/touch, low vision, and cognitive needs.
WCAG 2.2	Oct 5, 2023 (updated Dec 12, 2024)	**The current W3C Recommendation** — the version W3C tells you to use.
WCAG 3.0 ("Silver")	In progress	**Working Draft only** (last updated March 2026). Not a standard and not required.

WCAG 2.2 adds **9 new success criteria** over 2.1 (covering clearer keyboard focus, easier interactions such as no forced dragging, minimum target sizes, and less painful logins) and **removes** the old "4.1.1 Parsing" criterion, which modern browsers made obsolete. Versions 2.0, 2.1, and 2.2 are **backward compatible** — meeting 2.2 means you also meet the earlier versions.

> **WCAG 3.0 is not something to comply with today.** It is an early Working Draft that will not be a finished standard for a few more years. It aims for a broader scope (not just web content but apps, tools, and publishing), a more flexible scoring-based conformance model to replace the rigid A/AA/AAA system, and much stronger coverage of cognitive disabilities. *Flag:* many third-party articles describe WCAG 3 using "Bronze/Silver/Gold" tiers; earlier drafts used that idea, but the current official W3C page does not confirm those tier names, so treat the conformance model as **not finalized**. WCAG 2.2 AA will remain the operative target for years.

The rest of the WCAG family

- **WAI-ARIA (ARIA)** — "Accessible Rich Internet Applications." A specification that adds **roles, states, and properties** to HTML so screen readers can understand dynamic, custom widgets (menus, tabs, sliders, modals). Rule of thumb: use native HTML first; use ARIA only when native elements cannot do the job.
- **ATAG** — Authoring Tool Accessibility Guidelines. Covers the tools that *create* content (CMSs, editors like WordPress).
- **UAAG** — User Agent Accessibility Guidelines. Covers browsers, media players, and assistive tech.

What "compliance" actually means — and a warning for engineers

Conformance is WCAG's own concept: a page conforms at a level only if it satisfies *all* the success criteria at that level (and below), and full processes (like a checkout) must be accessible end-to-end. **Compliance** is the legal/contractual version: meeting whatever standard a law or contract names, almost always WCAG 2.1 AA or 2.2 AA.

> **Critical for an AI/ML team:** conformance is **not fully automatable**. Automated scanners catch only part of the picture (see Q4), and there is **no official W3C "certification."** "Passed a scanner" is not the same as "conformant." Human and assistive-technology testing and judgement are always required for a real conformance claim. This single fact shapes where AI helps and where it over-promises.

3. Q2 — Why was it introduced, and why should an owner care?

The origin and the vision

The web's inventor, **Tim Berners-Lee**, stated the founding principle plainly: *"The power of the Web is in its universality. Access by everyone regardless of disability is an essential aspect."* The web was conceived as a universal medium; accessibility is part of that original vision, not an add-on.

To make that real, the W3C launched the **Web Accessibility Initiative (WAI) in 1997**, with the explicit goal of **removing the barriers** that exclude people with disabilities. WCAG 1.0 followed in 1999, 2.0 in 2008, and the standard has evolved since. The purpose has never changed: **equal access to information and functionality for everyone**.

"Why should I, as an owner, actually care?" — the legal case

Beyond it being the right thing to do, there is real, growing legal exposure. As of 2026:

United States

- **ADA (Americans with Disabilities Act).** The ADA has no explicit "web" text, but courts and the U.S. Department of Justice (DOJ) have long treated the websites and apps of covered businesses as subject to it. This drives the bulk of U.S. litigation.
- **DOJ Title II Final Rule (April 2024)** — a landmark. It requires **state and local governments** to make their websites and mobile apps conform to **WCAG 2.1 Level AA**. *Important 2026 update:* in April 2026 the DOJ issued an interim rule **extending the deadlines by one year** — now **April 26, 2027** for larger entities (populations of 50,000+) and **April 26, 2028** for smaller ones. The DOJ stressed this is **not a pause** on the underlying obligation. (Separately, a U.S. HHS rule set a **May 11, 2026** deadline for many federally funded healthcare organizations.)
- **Section 508** of the Rehabilitation Act requires U.S. **federal agencies and their vendors** to make technology accessible, referencing WCAG 2.0 AA. This makes accessibility a **procurement requirement** — sell to the U.S. government and you must conform.
- **ACAA** (Air Carrier Access Act) governs airline websites and kiosks.

> **The tangible risk in numbers:** according to UsableNet's tracking, U.S. digital-accessibility lawsuits have run at record levels — **more than 5,000 filed in 2025**. Roughly **70% target e-commerce**, and about **46% of federal cases hit companies that had already been sued** — meaning genuine remediation measurably reduces repeat risk.

European Union

- **European Accessibility Act (EAA), Directive (EU) 2019/882.** Enforcement **began June 28, 2025**. It requires a broad range of **private-sector** products and services to be accessible — e-commerce, banking, e-books, ticketing and self-service terminals, smartphones, e-communications, and transport. Compliance is met through the harmonized standard **EN 301 549**, which currently maps to **WCAG 2.1 AA**. Penalties vary by country (up to roughly €900,000 in some), and crucially **any company selling**

into the EU is in scope, including U.S. and global businesses. Very small microenterprises providing services, and a documented "disproportionate burden" defence, are the main exemptions.

United Kingdom

- **Equality Act 2010** requires "reasonable adjustments," widely understood to cover websites, plus a separate public-sector accessibility duty.

The business case (the reasons beyond fear of lawsuits)

- **Market size.** The **WHO** estimates **1.3 billion people — about 16% of the world, roughly 1 in 6 — live with a significant disability.** Excluding them excludes a huge audience.
- **Spending power.** The disability market (people with disabilities plus their friends and family) is estimated by the Return on Disability Group to command roughly **$13 trillion** in annual disposable income; the UK's "Purple Pound" is put at around **£274 billion**. Users routinely abandon inaccessible sites and spend elsewhere. (These are advocacy/industry estimates — directionally reliable, not audited figures.)
- **SEO overlap.** Accessibility and search-engine optimization overlap heavily: semantic HTML, descriptive alt text, real text (not text in images), captions/transcripts, and logical headings help *both* screen readers and search crawlers.
- **Better UX for everyone.** Captions help in noisy or silent settings; high contrast helps in sunlight; keyboard operability helps power users; plain language helps non-native speakers.
- **The aging population & mobile.** Age brings vision, hearing, motor, and cognitive change, so the accessible audience keeps growing; and many WCAG requirements (target size, contrast, reflow) directly improve the mobile experience that is now most of web traffic.
- **Cheaper when built in.** Retrofitting accessibility after launch costs far more than designing it in — the core argument for building it into engineering from the start.

4. Q3 — What areas does accessibility touch and apply to?

WCAG is deliberately broad. It applies to essentially **anything delivered as digital content or interface**, across industries and content types. Think of it in two layers: the *domains/platforms* it covers, and the *content types* you actually fix within them.

Domains and platforms

Area	Why it's in scope / what's special
Websites	The original and most common target (marketing sites, blogs, portals).
Web applications	SaaS dashboards, single-page apps. Harder because of dynamic content and custom widgets — ARIA and focus management are essential.
Mobile apps	iOS/Android. WCAG principles apply; platform APIs (Apple VoiceOver, Android TalkBack) are the implementation layer. A growing source of lawsuits.
PDFs & documents	PDFs, Word, PowerPoint must be tagged and structured (reading order, headings, alt text, real text not scanned images). A very common failure point (related standard: PDF/UA).
E-commerce	The single most-litigated sector (~70% of U.S. cases). Product pages, search/filters, cart, and checkout must all be operable end-to-end.
Government / public sector	Bound by Section 508 (US federal), the DOJ Title II rule (US state/local), the EU Web Accessibility Directive, and UK public-sector rules.
Education / e-learning	LMS platforms, course content, quizzes, video lectures (captions/transcripts), PDFs.
Banking / finance	Online banking, statements, apps — explicitly named in the EAA; a common litigation target.
Healthcare	Patient portals, appointment booking, telehealth — ADA plus the U.S. HHS rule with a May 11, 2026 deadline.
Media / streaming	Captions (deaf/hard-of-hearing), audio description (blind/low-vision), and accessible players/controls.
Kiosks, ATMs, IoT	Ticketing machines, check-in kiosks, point-of-sale, connected devices — explicitly covered by the EAA / EN 301 549.
Email & SaaS/software	HTML emails need structure, alt text, contrast; enterprise software increasingly needs a VPAT (a conformance report) to be purchased.
Gaming	An emerging frontier: remappable controls, captions, colourblind modes, assist options (supplementary Game Accessibility Guidelines are used).

Content types and interaction patterns (the “what” you fix)

Text (readable, resizable, good contrast, real text, plain language); **images** (meaningful ones need alt text, decorative ones are hidden, complex charts need long descriptions); **audio** (needs transcripts); **video** (needs captions, often audio description and transcripts, no surprise autoplay); **forms** (properly associated labels, clear instructions, accessible error messages); **navigation** (consistent menus, skip links, descriptive links); **colour & contrast** (meet minimum ratios; never use colour alone to convey meaning); **keyboard access** (everything operable without a mouse, no traps, visible focus); **focus management** (correct focus movement when modals and dynamic content open/close); and **motion & timing** (adjustable time limits, pause/stop for moving content, and nothing flashing more than three times per second because of seizure risk).

> **Bottom line:** if it renders in a browser, on a phone, on a screen, or as a downloadable document, WCAG applies — and the same POUR principles govern every content type within it, from a one-page brochure to a global banking platform.

5. Q4 — Current best approaches used in industry *without* AI

Before adding AI, it helps to know what "good" looks like today. Everything in this section is the established, human-driven, **deterministic** discipline of accessibility engineering. Importantly, the well-known tools here (axe-core, WAVE, Lighthouse, Pa11y, ESLint plugins) are **rule-based engines, not AI**: they run a fixed catalogue of coded checks against the page and return the same result every time. That reproducibility is exactly why they are trusted in compliance workflows. The "intelligence" in accessibility today comes from standards, human expertise, and testing with real users.

Foundational engineering practices

Getting these right resolves the large majority of defects before any tool runs:

- **Semantic HTML first.** Use the correct native element (`<button>`, `<a>`, `<nav>`, `<h1>`–`<h6>`, `<label>`). Native elements bring keyboard behaviour, focus handling, and screen-reader meaning for free. This is the single highest-leverage practice.
- **Document structure & landmarks.** A logical heading hierarchy and landmark regions let screen-reader users jump around by structure. Provide a "skip to main content" link.
- **WAI-ARIA — used sparingly.** ARIA communicates custom widgets to assistive tech, but wrong ARIA actively breaks things, hence the community maxim **"No ARIA is better than bad ARIA."** The Five Rules of ARIA: use native HTML first; don't override native semantics; make all interactive ARIA keyboard-operable; never hide a focusable element; and give every interactive element an accessible name.
- **Keyboard access & focus management.** Everything operable by keyboard, no traps, logical tab order, visible focus indicators (tightened in WCAG 2.2), and correct focus movement in modals and single-page-app route changes.
- **Colour contrast.** At least 4.5:1 for normal text, 3:1 for large text and UI components; never rely on colour alone.
- **Text alternatives & accessible forms.** Descriptive `alt` for meaningful images, empty `alt` for decorative ones; every input has an associated label; errors are described in text and tied to the field.
- **Responsive, zoom & reflow, and motion controls.** Content reflows to one column at 400% zoom, text resizes to 200%, moving content can be paused, nothing flashes more than three times a second, and the `prefers-reduced-motion` setting is honoured.

Process best practice: "shift left"

The dominant methodology is **shift-left** — handle accessibility as early as possible instead of remediating at the end (industry playbooks cite roughly 75% cost savings versus late fixes). In practice that means: accessible design systems and Figma annotations (reading order, focus order, alt text) in the **design** phase; contrast baked into design tokens; reusable accessible components and editor linters in **development**; automated + manual gates in **QA**; and an explicit **accessibility "Definition of Done"** on every ticket so a feature isn't "done" until it is keyboard-operable, labelled, contrast-checked, and screen-reader tested. POUR doubles as the everyday checklist.

Manual testing & auditing — the gold standard

> **The central, well-established fact of the field:** automated tools alone are not enough. The W3C states it plainly — *"no tool alone can determine if a site meets accessibility standards… knowledgeable human evaluation is required."*

How much do automated tools catch? Two honest framings:

- **By criteria:** roughly **16 of the ~50** WCAG 2.1 AA success criteria are machine-testable (about 32%) — the source of the familiar "automation catches only about a third" claim.
- **By issue volume:** Deque's Automated Accessibility Coverage Report re-measured this across ~300,000 real issues and found automation surfaces about **57%**, leaving roughly **43%** for humans — because a few auto-detectable issues (like contrast) are extremely frequent.

Either way the conclusion is the same: **automation is a first pass, not a verdict.** The rest requires **expert manual audits** against each success criterion, **testing with real assistive technologies** (per WebAIM's 2024 survey the common matrix is JAWS, NVDA, and VoiceOver screen readers, plus magnifiers, voice control, and switch access), **keyboard-only testing**, and — the highest-fidelity signal — **usability testing with people with disabilities.**

The traditional (non-AI) automated tools

Tool	What it is
axe-core (Deque)	The open-source rules engine that is the de-facto industry standard and powers many other tools. Deliberately minimizes false positives; flags uncertain cases as "needs review."
WAVE (WebAIM)	Visual in-browser evaluation that overlays icons on the page — great for learning and assisted manual review.
Lighthouse (Google)	Built into Chrome DevTools; its accessibility audit is a subset of axe rules.
Pa11y	Command-line/CI runner for automated pipelines and dashboards.
eslint-plugin-jsx-a11y	Static code checker that catches issues in React/JSX at authoring time, before the app even runs — shift-left at the linter level.

The mature pattern is to run the linter in the editor, run axe-core in the test suite, and **fail the build on new violations** — always paired with periodic manual audits.

Documentation & governance

A **VPAT** (Voluntary Product Accessibility Template) filled in becomes an **Accessibility Conformance Report (ACR)**, which documents, criterion by criterion, how a product conforms (Supports / Partially Supports / Does Not Support / Not Applicable). ACRs are standard in procurement. Organizations also publish **accessibility statements**, run structured site audits with the W3C's **WCAG-EM** methodology, maintain prioritized **remediation roadmaps**, and run standing accessibility programs with training and a **maturity model**. The common target is **WCAG 2.1/2.2 AA**, aligned to **EN 301 549** in Europe, with the **ARIA Authoring Practices Guide** as the reference for building components correctly.

6. Q5 & Q6 — Where AI can be utilized (7 use cases + the players who built them)

This is the heart of the brief. Below are seven high-impact places where AI is already being used to make accessibility more efficient, accurate, faster, and user-friendly. Each is written in the exact presentation format your team specified. Immediately after each use case, the **existing players** (Q6) are profiled in depth — what the product does, how its AI works, and a direct link you can open to review it yourself.

> **How the seven were chosen.** Per your scoping decision, this catalogue focuses on the **7 highest-impact, most production-relevant** use cases rather than an exhaustive list, and profiles the **2–3 leaders** in each. They are ordered roughly from most mature/lowest-risk to most emerging.

#	Use case	Maturity
1	AI-Generated Alt Text & Image Descriptions	Mature, shipping
2	AI-Powered Automated Accessibility Testing & Auditing	Production (as augmentation)
3	Automatic Captioning, Transcription & Audio Description	Mature (captions); emerging (AI audio description)
4	AI Accessibility Overlays / Automated Remediation Widgets	Deployed but controversial
5	AI Sign-Language Translation & Avatars	Constrained production + active research
6	AI Cognitive Accessibility — Text Simplification & Reading Support	Mature, low-risk
7	AI Conversational / Voice Interfaces & Visual Assistance	Production (assistants); emerging (agents)

Use Case 1 — AI-Generated Alt Text & Image Descriptions

Name of the use case / area

AI-generated alternative text and rich image descriptions for images that ship with missing, empty, or useless alt text.

Detailed description

A very large share of web images have no useful alt text, which makes them invisible to blind and low-vision users on screen readers. AI closes this gap by generating a natural-language description of an image on demand — at authoring time (alt-text suggestions inside a CMS or Office document), at browse time (the browser or OS describes an unlabelled image), or interactively (a user photographs their surroundings or a screen and asks follow-up questions). Modern systems have moved from one-line "captions" to rich, conversational, multi-paragraph descriptions with question-and-answer follow-ups.

Tech stack to build the use case

Vision-Language Models / VLMs (GPT-4o, Claude, Gemini) for rich conversational descriptions; dedicated image-captioning models (e.g., Microsoft's visual-vocabulary caption model); supporting computer-vision primitives — OCR/text recognition, object detection, scene/image classification; cloud APIs such as Azure AI Vision (Image Analysis 4.0), Google Cloud Vision, and OpenAI vision endpoints; plus text-to-speech (TTS) for spoken output on mobile.

Detailed approach to solve the use case

Two dominant patterns. **(1) Purpose-built caption models** (Microsoft, Google), trained on large image–text and image–tag datasets and gated on a confidence threshold so weak guesses are suppressed. **(2) VLM-as-a-service** (the Be My AI approach): send the image (and optional user prompt) to a hosted multimodal model, return a long-form description, and support multi-turn Q&A — minimal in-house modelling, with product value in UX, safety guardrails, and a human fallback. The differentiator across all of them is **hallucination handling and human-in-the-loop**: Google only surfaces a description when confident and errs toward saying less or nothing; Be My Eyes keeps its volunteer network one tap away for verification; Microsoft treats auto-captions as editable suggestions, used only when a human-written description is missing.

Complexity level LOW–MEDIUM to integrate HIGH to build a frontier model

Calling a hosted VLM or Vision API for captions is genuinely low-effort (a few API calls plus TTS). Building a caption model at human parity is high-complexity research. The hard *product* problems are hallucination control, latency, cost at web scale, and describing context the pixels alone do not contain.

Pros and cons

Pros	Cons
Massive coverage of otherwise-undescribed images; near-instant; conversational follow-up answers user-specific questions; increasingly available on-device; cheap to integrate via APIs.	Hallucinations (a confidently wrong description is worse than none for someone who can't verify it); the AI lacks the author's intent and page context; privacy of user-photographed content; cost/latency at scale; coverage is English-first.

Buildable with existing AI tech? **Yes — the most production-ready of the seven.** This is essentially a hosted-model integration problem today.

Existing players (Q6)

Be My Eyes — “Be My AI”

A free app for blind/low-vision users: take or share a photo, get a detailed description, then ask follow-up questions in chat. Also deployed inside enterprise customer-service centres as an image-based support channel. **How the AI works:** powered by OpenAI’s multimodal GPT-4 / GPT-4o; Be My Eyes builds the accessible UX and safety layer rather than training its own vision model. **Human-in-the-loop:** one-tap escalation to its network of sighted volunteers when the AI is unsure or the user wants to verify.

https://www.bemyeyes.com/ • Introducing Be My AI

Microsoft — Azure AI Vision captioning + Seeing AI

Azure AI Vision (Image Analysis 4.0) produces captions and region-level “dense captions.” It powers auto alt-text suggestions in Word, Outlook, and PowerPoint, and the free **Seeing AI** mobile app that narrates the physical world for blind users. **How the AI works:** a “visual vocabulary” pre-training step on image/word-tag pairs followed by caption fine-tuning; Microsoft reports reaching or surpassing human parity on the industry *nocaps* novel-object benchmark. Auto-captions are surfaced as editable suggestions for a human author.

Azure image captioning • Azure AI Vision docs

Google — “Get image descriptions from Google” (Chrome / ChromeOS)

Built into Chrome for screen-reader users: an opt-in setting that auto-describes unlabelled web images. **How the AI works:** an ensemble of text recognition (OCR), object detection, and a scene-description model runs on Google servers, evaluates output for accuracy/usefulness, and only surfaces a description when reasonably confident — explicitly erring toward a simpler answer or nothing. It has generated many millions of descriptions. Google is candid that machine descriptions are “not as good as a human.”

Google announcement • User help

Also in this space (not deep-profiled): Meta automatic alt text, CloudSight (commercial image-recognition API), and Scribely (a human-expert alt-text service — a notable “human-first” counterpoint).

Use Case 2 — AI-Powered Automated Accessibility Testing & Auditing

Name of the use case / area

Using machine learning and computer vision to detect accessibility problems that deterministic rule-based scanners structurally cannot judge.

Detailed description

Rule-based scanners (axe-core, WAVE, Lighthouse) reliably catch only the machine-checkable issues — a missing `alt` attribute, a low contrast ratio, a missing form label. They cannot judge *meaning*: is this alt text actually descriptive or is it "image123.jpg"? Is this bold text a real heading or just styled text? Is this `<div onclick>` really a button? AI adds a semantic/visual layer — computer vision that "sees" the rendered page like a sighted user, ML that predicts whether content is *meaningful*, and guided workflows that walk a human through only the subjective checks that automation can flag but not decide.

Tech stack to build the use case

Computer vision on rendered screenshots (element detection, spatial/semantic relationships, visual OCR); ML classifiers for "is this a real heading / button / meaningful label"; semantic modelling combining the accessibility tree + visual render + rule engine; DOM-mutation tracking to test dynamic single-page-app states; a deterministic core (axe-core rules) as the trusted base layer; and increasingly LLM assistants for fix suggestions and IDE integration.

Detailed approach to solve the use case

The pattern is **hybrid: deterministic rules + an ML overlay**. Render the page in a real browser; capture the DOM, accessibility tree, and pixels. Run axe-core-style rules for the certain issues. Then layer CV/ML to (1) find interactive or visual elements that lack correct semantic code, (2) extract visible text and compare it with what assistive tech would announce, and (3) route genuinely subjective items into **guided testing** — a wizard that presents a human tester with just the elements needing a judgement call (data tables, headings, alt-text quality) and records the decision. The leaders frame this as "automation assists, humans decide," and treat false positives as bugs, because noisy reports destroy developer trust.

Complexity level HIGH

Requires a headless rendering farm, CV models trained and validated on UI elements, robust crawling of dynamic app states, and — critically — keeping false positives near zero. Getting ML to reliably judge "is this alt text meaningful" is genuinely hard and still human-assisted.

Pros and cons

Pros	Cons
Catches issues rule engines structurally cannot (visual/semantic mismatches, unlabelled custom widgets); slashes manual-audit time; tests dynamic states; scales across huge sites and CI pipelines.	Still cannot fully replace expert manual + assistive-tech testing; ML brings its own false positives/ negatives; "meaningfulness" remains partly subjective; risk of over-claiming "AI compliance."

Buildable with existing AI tech? **Yes — as augmentation, not full automation.** Even the leaders keep humans in the decision loop; no credible tool claims fully automated WCAG conformance.

Existing players (Q6)

Deque — axe DevTools (ML + Intelligent Guided Testing)

The axe platform pairs the deterministic axe-core engine with an ML/computer-vision layer and **Intelligent Guided Tests** that walk testers through subjective checks (reportedly up to 4× faster than standard manual review). **How the AI works:** computer vision extracts semantic data from the rendered page — analysing interface elements, spatial relationships, and visual text — to feed the guided tools; newer additions include AI fix suggestions, an "Axe MCP Server" delivering remediation guidance inside the IDE, and a chatbot trained on Deque University content. Philosophy: automation assists, humans decide, targeting zero false positives.

https://www.deque.com/axe/ai/ • ML in axe (blog)

Evinced — computer-vision accessibility engine (web + mobile)

A developer-focused platform that "renders pages and analyses them like a sighted user would," with native SDKs for iOS and Android. **How the AI works:** advanced rule-sets plus computer vision and AI build a *structural semantic model* of the page, detecting mis-coded interactive elements (e.g., a `<div>` acting as a button) that static checkers miss, and automatically tracking DOM mutations from user interactions to test dynamic states — framework-independent. Evinced claims it surfaces roughly 19× more critical issues than legacy tools with fewer false results.

https://www.evinced.com/technology

Siteimprove — AI-assisted testing at scale

An enterprise platform that crawls whole sites, scores accessibility, and offers an AI-assisted Accessibility Code Checker. Positioned more as site-wide governance, monitoring, and trend reporting than a computer-vision-first detection engine. (Partially verified: ACR page and Code Checker confirmed; specific ML internals are not publicly disclosed.)

https://www.siteimprove.com/acr/

> **What AI adds beyond axe-core:** deterministic rules answer "is the attribute present and syntactically valid?" The AI layer attempts "is it *meaningful and correct* for a real user, and is this visually-a-control actually a coded control?" — which is the bulk of real-world WCAG failures.

Use Case 3 — Automatic Captioning, Transcription & Audio Description

Name of the use case / area

AI speech-to-text for captions and transcripts, and AI-generated audio description of video for blind users.

Detailed description

Video and audio must be perceivable to deaf and hard-of-hearing users (captions, transcripts) and to blind users (audio description of purely visual content). AI provides automatic speech recognition (ASR) for captions and transcripts — both live and batch — plus speaker labelling, punctuation, and translation; and, at the frontier, **AI audio description**, where a vision-language model describes on-screen visuals and TTS narrates them into the natural pauses of a video. This is timely: the U.S. ADA Title II deadlines are pushing public universities and governments to caption huge back-catalogues of video.

Tech stack to build the use case

ASR/speech-to-text (Whisper-class models, Azure AI Speech, Google on-device ASR, Amazon Transcribe, Speechmatics); punctuation, casing, and speaker-diarization models; language identification and translation for subtitles; a vision-language model plus scene analysis and TTS for audio description; human-review tooling (transcript editors, accuracy prediction); and streaming infrastructure for live captions.

Detailed approach to solve the use case

Captions/transcripts follow an **ASR → post-processing (punctuation, casing, diarization) → optional human correction** pipeline. The key production lever is **confidence-based routing**: segments above a quality threshold ship AI-only, while weaker ones route to human editors — letting vendors reach the 99%+ accuracy that legal and education compliance demand at manageable cost. **Live captioning** runs streaming ASR with interim (partial) results for low latency; Google's Live Caption does this fully on-device for privacy and speed. **AI audio description** is newest: detect the visual gaps, generate descriptions with a VLM, fit them into dialogue pauses, and synthesize speech.

Complexity level **MEDIUM (captions/transcripts)** **HIGH (live + audio description)**

Batch captioning from a hosted ASR is very approachable. Real-time low-latency captioning, domain accuracy (legal/medical terms, accents), speaker labelling, and especially good audio description (timing into pauses, deciding what is worth describing) are hard.

Pros and cons

Pros	Cons
Orders of magnitude cheaper and faster than manual captioning; scales to giant back-catalogues; on-device options preserve privacy; live captions can appear anywhere.	ASR errors on names, jargon, accents, and crosstalk; compliance often needs ~99% accuracy that pure AI misses (so human review is required); AI audio-description quality is still unproven versus professional describers; live captions can lag and garble.

Buildable with existing AI tech? **Yes for captions and transcripts (mature).** AI audio description is real but **emerging** and not yet at professional-human parity.

Existing players (Q6)

3Play Media — AI-enabled captioning + patented AI audio description

Captioning, transcription, subtitling, and audio description at scale, explicitly targeting the 2026 ADA Title II deadline for higher education; also live classroom captioning that can toggle between a verbatim transcript and an AI-generated lecture summary. **How the AI works:** ASR plus 15+ years of ML, with **predicted-accuracy thresholds** that route files to human review versus AI-only, and a **patented automated audio-description** pipeline. Expert human reviewers verify and refine output.

https://www.3playmedia.com/news/launches-ai-enabled-solutions/

Verbit — adaptive AI + professional captioners

Live and recorded captioning/transcription (including CART), plus translation, subtitling, and dubbing in 50+ languages, verticalized for legal, education, media, and corporate. **How the AI works:** a proprietary *adaptive* ASR engine customizable per industry/organization for accuracy on domain jargon, with real-time interim results for live delivery, backed by a network of professional human transcribers who correct output for high-stakes work — the hybrid model is the core positioning.

https://verbit.ai/

Google — Live Caption (on-device) & YouTube auto-captions; Microsoft — Azure AI Speech

Google Live Caption captions any audio on Android and in Chrome in real time using three on-device deep nets (RNN-T ASR + a punctuation net + a sound classifier for `[MUSIC]` / `[APPLAUSE]`), running offline for privacy and low latency; YouTube auto-captions billions of videos. **Microsoft Azure AI Speech** provides real-time and batch speech-to-text, TTS, and translation, and powers Teams and Stream live captions with custom speech models for domain accuracy.

Google Live Caption • Azure AI Speech

Also in this space: Otter.ai (meeting transcription), Rev (AI + human), Amazon Transcribe (ASR API), and OpenAI Whisper (open-weights ASR widely used as an in-house building block).

Use Case 4 — AI Accessibility Overlays / Automated Remediation Widgets

> **Read this one carefully.** This use case is deployed widely but is **seriously contested**. The report presents the vendor pitch and the expert criticism side by side, because a U.S. regulator has already penalized a leader for over-claiming. Bottom line up front: overlays do **not** achieve WCAG conformance on their own and can actively harm assistive-technology users.

Name of the use case / area

A single line of JavaScript added to a site that claims to auto-fix accessibility at runtime using AI (add alt text, fix ARIA, adjust contrast) plus an end-user preferences menu.

Detailed description

A vendor script is added to a website; at page load it scans the DOM and accessibility tree and claims to auto-fix issues — inject missing alt text via image recognition, correct ARIA roles and labels, repair headings and form associations — and it exposes a widget where end users can change contrast, font size, and motion. It is marketed as fast, cheap "AI compliance" without touching source code.

Tech stack to build the use case

Client-side JavaScript injected at page load, manipulating the DOM/accessibility tree; image recognition/OCR for alt text; ML/NLP to guess ARIA roles and element semantics; a user-facing widget UI (contrast, text size, profiles); and vendor-side scanning/monitoring dashboards. Hybrid vendors add human audit tooling on top.

Detailed approach to solve the use case

Overlays run entirely at runtime in the browser after the page loads, mutating the accessibility tree rather than fixing the source. The most defensible variant is **hybrid**: **AudioEye** uses automation for the issues it can fix instantly, then has human experts and disabled testers manually test with real assistive tech and deploy custom fixes through the same JavaScript framework. **UserWay** similarly layers code-level fixes, monitoring, expert audits, and VPAT/ACR services on top of the base widget. **accessiBe's** accessWidget relied heavily on fully-automated AI remediation — the exact approach a regulator found could not deliver its central promise.

Complexity level LOW to deploy — the "compliance" claim is effectively unsolved

Dropping in a snippet is trivial. But reliably repairing accessibility at runtime — correct ARIA on custom widgets, meaningful alt text, keyboard operability in modern app frameworks — is precisely what experts and regulators say automation cannot do. The easy part is deployment; the hard part is the honesty of the claim.

Pros and cons

Pros (mostly vendor-claimed)	Cons (expert consensus)
One-line install; instant partial fixes; low cost; the contrast/text-size user tools genuinely help some low-vision users; hybrid vendors add real human remediation and legal support.	The **Overlay Fact Sheet** — signed by 1,000+ accessibility experts, including contributors from Google, Microsoft, Apple, and the NVDA/JAWS screen readers — states overlays **cannot achieve WCAG conformance**. They can **break screen readers** (a WebAIM survey found ~67% of assistive-

	tech users rate overlays “not at all/not very effective”); automated repair of alt text, labels, and keyboard behaviour is unreliable; the widget controls duplicate what the OS/assistive tech already provide; and sites using overlays have **themselves been sued**, undercutting the “legal protection” pitch.

> **The accessiBe FTC action (verified).** On **January 3, 2025** the U.S. Federal Trade Commission ordered accessiBe to pay **$1,000,000**, finding it falsely claimed its accessWidget “could make any website compliant with WCAG,” and that it failed to disclose paid connections to supposedly independent reviewers. The FTC **approved the final order on April 22, 2025**, barring the company from claiming automated products ensure WCAG compliance without evidence. This is the clearest signal that over-claiming automated accessibility is now a real regulatory liability.

Buildable with existing AI tech? **Partly — and that is the trap.** An AI overlay is buildable and deployable today; what is *not* achievable with current tech is the marketed promise of automated, full WCAG conformance. The honest ceiling: overlays can provide some user-preference conveniences and a subset of automated fixes, but real conformance still requires source-level remediation and expert/assistive-tech testing.

Existing players (Q6)

AudioEye — hybrid AI + human (the most defensible model)

Operates on the DOM/accessibility tree; automation fixes a portion of issues instantly, then human experts and disabled testers handle complex components, ambiguous semantics, and nuanced alt text. Its own framing: “a widget changes what a sighted user sees; AudioEye changes what assistive technology reads.” The most credible of the three because it openly acknowledges automation’s limits and adds real human remediation — though it is still overlay-delivered and is named in the Overlay Fact Sheet.

How AudioEye works

UserWay — AI widget + tiered expert services

An AI-powered widget runs on every page load (contrast, alt text, ARIA, keyboard nav, user profiles). Beyond the widget, UserWay sells code-level fixes, monitoring, expert manual audits, VPAT/ACR documentation, video remediation, and litigation support — itself an acknowledgement that the widget alone is not conformance. Also named in the Overlay Fact Sheet.

https://userway.org/

accessiBe — accessWidget (cautionary case)

The category's best-known overlay, historically reliant on fully-automated AI remediation. Subject of the **$1M FTC settlement (2025)** for deceptive automated-compliance claims. Included here as the cautionary example of what over-claiming looks like, not as a recommendation.

Overlay Fact Sheet (expert criticism) • FTC order (Jan 2025)

Use Case 5 — AI Sign-Language Translation & Avatars

Name of the use case / area

Translating text/speech into sign language via animated or generative avatars, and recognizing sign language via computer vision.

Detailed description

Many Deaf people whose first language is a signed language (BSL, ASL, Libras, and others) treat written text as a *second* language, so captions and plain text alone do not guarantee comprehension. Two AI directions address this: **text/speech → sign** via animated 3D avatars or photorealistic generated signers (for websites, kiosks, and transport displays), and **sign → text/speech** via computer-vision recognition, so a Deaf signer can "type" or converse with hearing people and systems.

Tech stack to build the use case

Recognition (sign→text): keypoint/skeleton tracking (MediaPipe/OpenPose-style), optional depth/point-cloud sensors, CNN + temporal models (LSTM/Transformer) over keypoint sequences, and machine-translation-style decoding. *Synthesis (text→sign):* an LLM or neural machine translation for text→"gloss" (a written encoding of sign grammar), then either rigged 3D avatars (animation blending over a motion-captured sign library) or generative video (GAN/diffusion/neural rendering over a studio-recorded clip corpus with interpolation). Plus ASR for speech input, TTS for output, and large annotated sign corpora captured in multi-camera studios.

Detailed approach to solve the use case

Data is the bottleneck: high-quality annotated multi-view sign corpora are scarce and expensive to record. Pipelines are almost always **two-stage** (text↔gloss↔motion/video) because direct end-to-end text→signed-video is still immature. Human-in-the-loop — Deaf reviewers and advisory boards — is standard and effectively mandatory for both quality and community trust.

Complexity level HIGH

Signed languages are full natural languages with spatial grammar, simultaneity, and heavy reliance on facial and other non-manual features that current avatars render poorly. Both recognition (continuous, co-articulated, signer-variable) and synthesis (natural facial grammar, role shift) remain unsolved at human-interpreter quality, and data scarcity compounds it.

Pros and cons

Pros	Cons
Scales to contexts with no interpreter available (transport screens, 24/7 web content); consistent, on-demand, and cheaper per use than a live interpreter; genuinely serves sign-first Deaf users better than text alone.	The World Federation of the Deaf and WASLI have formally cautioned that avatars **cannot yet match human interpreters** — machine translation misses context and culture, and avatars struggle with the facial/non-manual expressions that carry grammatical meaning. Their position: acceptable for constrained, pre-scripted content with Deaf oversight, but not a replacement for interpreters in dynamic or high-stakes settings.

Buildable with existing AI tech? **Partially.** Constrained, template-driven text→sign (transport announcements, scripted web content) is in production now. Open-domain, real-time, interpreter-quality bidirectional translation is **not** achievable today; continuous conversational sign→text at scale is still research-grade.

Existing players (Q6)

Signapse (UK) — generative photorealistic sign-language video

Generative-AI sign-language translation delivered as photorealistic video of a human-looking signer (not a cartoon avatar). Products include SignStudio (translate web/video content), a real-time SignStream API, and automatic transport announcements — e.g., Cincinnati/CVG Airport used Signapse for live ASL flight updates. **How the AI works:** a two-stage pipeline — an LLM translates English into BSL/ASL "gloss," then computer-vision/generative models turn gloss into video by blending clips from a library of ~12,000 glosses recorded in a 30-camera studio. For transport, AI blends pre-approved clips and swaps in live data (times, gates). ~1/3 of staff are BSL users, with a Deaf Advisory Board reviewing translations.

https://www.signapse.ai/ • How the AI works

Hand Talk (Brazil) — 3D-avatar translation app & plugin

A mobile app and website plugin using the 3D avatars "Hugo" and "Maya" to translate text/speech into Libras (Brazilian Sign Language) and ASL; reports 1B+ words translated and 10M+ downloads, with a signed dictionary, adjustable speed, and offline mode. **How the AI works:** automatic machine translation from text into a signed-gloss sequence, rendered by a rigged 3D avatar. *Flag:* the exact ML architecture is proprietary and not publicly documented. The 3D-avatar approach is lighter-weight and cheaper to scale than generative video, at the cost of realism and facial-expression fidelity.

https://www.handtalk.me/en/app/

SignAll (US/Hungary) — sign→text via computer vision

The reverse direction: recognizing ASL and converting it to text (and optional TTS), built on a Gallaudet University partnership. **How the AI works:** one or more cameras (historically with depth sensors) drive skeleton/keypoint tracking of hands, fingers, body, and face; the system extracts the linguistic parameters of each sign (handshape, orientation, movement, location, and non-manual markers) and matches them against a large ASL vocabulary database using machine-translation-style probability scoring, surfacing top candidates for the signer to confirm.

https://www.signall.us/technology

Research/pilots worth watching (not production translators): Google Project Shuwa (real-time sign *detection* in video calls), SLAIT (pivoted from live translation to AI sign-language lessons in 2023), and academic LLM-over-sign research. The WFD/WASLI statement on signing avatars is the key community-perspective source.

Use Case 6 — AI Cognitive Accessibility: Text Simplification & Reading Support

Name of the use case / area

Using NLP/LLMs to simplify complex text, summarize, define words, read aloud, and personalize content for users with dyslexia, ADHD, cognitive or learning disabilities, aphasia, low literacy, or non-native readers.

Detailed description

Complex prose is a real barrier for a large population. AI helps by **rewriting text into plain language**, summarizing, providing on-demand word definitions and picture dictionaries, reading aloud (TTS) with word highlighting, breaking words into syllables and parts of speech, masking/line-focusing, and translating. LLMs are a genuine leap here: earlier tools could read text aloud or restyle it, but *reliable rewriting of arbitrary text into plain language* only became practical with modern LLMs. WCAG 3.0's greater emphasis on cognitive accessibility (still a Working Draft) points squarely at this use case.

Tech stack to build the use case

LLMs (GPT/Claude/Gemini-class) for simplification and summarization; NLP tokenizers, part-of-speech taggers, and syllabifiers; neural TTS for read-aloud; ASR for dictation; neural machine translation; OCR for reading text inside images and inaccessible PDFs; and front-end DOM/styling engines for masking, spacing, and theming.

Detailed approach to solve the use case

Two delivery patterns: **client-installed** tools (e.g., Read&Write) that overlay any app the user opens, versus **site-embedded** toolbars (e.g., Recite Me) that a publisher adds so any visitor can adapt the page. Simplification is typically **prompt-based LLM rewriting** toward a target reading level. The key engineering risk is **factual drift** — an LLM can subtly change meaning while "simplifying" — so responsible deployments constrain prompts and keep human review for high-stakes content. Personalization stores each user's preferences (voice, spacing, theme, reading level).

Complexity level LOW to MEDIUM

The plumbing (TTS/ASR/translation/OCR APIs, DOM styling) is commoditized and cheap. The genuinely hard part is **trustworthy simplification** — preserving meaning, tone, and correctness, and measuring readability gains without hallucination. So: low to build a toolbar, medium to make the AI simplification safe and reliable.

Pros and cons

Pros	Cons
Immediately useful with a strong evidence base for read-aloud + highlighting; LLM simplification opens content to a huge underserved population; almost entirely buildable on existing APIs.	LLM simplification can hallucinate or distort meaning; over-simplification can strip nuance; site-overlay tools are sometimes criticized as no substitute for accessible source content; personalization raises privacy considerations; "AI accessibility" claims can create false compliance confidence.

Buildable with existing AI tech? **Yes, fully.** All components are production-grade today. This is the most mature and lowest-risk of the emerging use cases; the main work is quality control on simplification.

Existing players (Q6)

Microsoft — Immersive Reader

A free reading engine embedded across Edge, Word, OneNote, Teams, and Outlook, and offered as an **Azure Cognitive Service API** developers can embed in their own products. Features include Read Aloud (adjustable voice/speed with word highlighting), Line Focus, a Picture Dictionary, translation into 100+ languages, adjustable text size/spacing/column width, and syllable splitting and parts-of-speech colouring. **How the AI works:** Azure neural TTS for read-aloud, neural machine translation for languages, and NLP tokenizers/POS taggers for syllables and grammar colouring — delivered as a reusable API, which is why it shows up inside so many third-party education products.

Immersive Reader

Texthelp / Everway — Read&Write & OrbitNote

The market-leading literacy toolbar for education and the workplace. Features include natural-voice TTS with word highlighting, a Screenshot Reader (OCR read-aloud of images and inaccessible PDFs), picture and text dictionaries, speech-to-text, word prediction, translation, screen masking, and highlight-collection with summarization — and now a “Simplify” feature that applies LLM-style text simplification. **How the AI works:** it combines classic assistive components (OCR, TTS, ASR, word-prediction models) with AI-powered simplification and realistic voice synthesis; the Simplify feature is the LLM leap layered onto a mature literacy stack.

https://www.everway.com/products/read-and-write-education/

Recite Me — site-deployed assistive toolbar

A toolbar that *site owners embed* so any visitor can adapt the page (in contrast to a user-installed tool), running on 12,500+ sites. Features include TTS that reads text and image alt text, translation into 100+ languages with 65+ voices, font/size/colour customization, a screen mask and reading ruler, audio download, a dictionary, and an “AI Simplified Content” feature that distils full pages into concise summaries. **How the AI works:** LLM-based summarization/simplification, plus neural machine translation and TTS, with the rest handled by client-side styling.

https://reciteme.com/product/assistive-toolbar/

Also worth naming: browser-native reading modes (Edge Immersive Reader, Safari Reader, Firefox Reader View) provide free read-aloud/line-focus baselines, and Helperbird is a popular browser-extension alternative.

Use Case 7 — AI Conversational / Voice Interfaces & Visual Assistance

Name of the use case / area

Voice-driven navigation, conversational agents as an accessible interface, and AI visual-interpreting assistants that describe the world for blind users — with LLM "computer-use" agents as the emerging frontier.

Detailed description

Three overlapping strands: (1) **voice-driven control** for users who cannot use a mouse or keyboard; (2) **conversational agents/chatbots** as an accessible front door to complex sites (ask instead of navigate); and (3) **AI visual-interpreting assistants** that describe images and surroundings for blind/low-vision users in conversation. The emerging edge is **LLM agents that navigate websites on a user's behalf**, effectively routing around inaccessible interfaces.

Tech stack to build the use case

ASR (Whisper-class); LLMs (GPT-4o/Claude/Gemini); multimodal vision-language models (GPT-4o, Gemini/Project Astra) for live visual understanding; neural TTS; conversational frameworks (Voiceflow, Rasa, Dialogflow); streaming/real-time infrastructure for live audio+video; and accessibility-tree/DOM interfaces for agentic navigation.

Detailed approach to solve the use case

Visual assistance: stream frame(s) → VLM → conversational response → TTS, with human escalation as a safety net. *Voice interfaces:* ASR → intent/LLM → action/API → TTS. *Agentic web navigation:* an LLM plans over the accessibility tree or rendered DOM and executes clicks and inputs. Human-in-the-loop remains central for high-stakes visual interpreting; hallucination and latency are the key engineering constraints.

Complexity level **MEDIUM (assistants/voice)** **HIGH (reliable autonomous agents)**

Building a GPT-4o-backed visual describer or a Voiceflow voice agent is very achievable today. Making an LLM agent *reliably and safely* transact on arbitrary websites for a vulnerable user is hard — the error tolerance is low and mistakes have real consequences.

Pros and cons

Pros	Cons
Large independence gains for blind/low-vision and motor-impaired users; on-demand, private, 24/7; conversational interfaces lower cognitive and navigation load; hybrid human+AI provides a safety net.	VLMs hallucinate (dangerous when describing medications, hazards, currency, or navigation); latency in live scenarios; privacy of camera/voice streams; AI is not a legal substitute for human interpreting in high-stakes settings; agentic navigation raises consent, security, and reliability risks.

Buildable with existing AI tech? **Yes for visual-assistance and voice agents** (shipping in production). **Partially for autonomous web-navigation agents** — real but not yet reliable or safe enough to be a mainstream accessibility solution.

Existing players (Q6)

Be My Eyes — “Be My AI” (conversational visual assistant)

Beyond one-shot alt text (Use Case 1), Be My AI is a **conversational** multimodal assistant: a blind user photographs anything — a fridge, a document, a screen, a street — and gets a rich description plus **multi-turn follow-up Q&A** (“what can I cook with this?”). **How the AI works:** OpenAI’s GPT-4o for image understanding and dialogue, with one-tap handoff to human volunteers when needed. It is the strongest example of an LLM as an accessible conversational interface.

https://www.bemyeyes.com/

Aira — human visual interpreting, now AI-augmented

Professional visual interpreting: trained human agents view a blind user’s live camera and describe surroundings, read mail, navigate airports, and complete forms — widely sponsored free at airports and campuses. In 2025 Aira added an **AI Visual Interpreter powered by Google DeepMind’s Project Astra (Gemini)** for real-time conversational assistance, with seamless escalation to a human agent for complex or high-stakes tasks. Aira is the clearest example of the industry direction: **human-first, AI-augmented**.

https://aira.io/ • AI Visual Interpreter

Voiceflow — the builder layer for accessible voice/chat agents

An enterprise no-code/low-code platform to build, deploy, and scale chat and voice AI agents — relevant as the toolkit for creating a conversational/voice interface that serves as an accessible alternative to visually complex navigation. **How the AI works:** model-agnostic (choose GPT, Claude, Gemini, or Llama), a visual flow builder for dialogue, and integrations with ASR/TTS providers and back-ends. (A general CX platform, not an accessibility-specific product, but a practical way to ship accessible voice/chat front-ends.)

https://www.voiceflow.com/

Platform voice assistants (Apple Voice Control, Amazon Alexa, Google Assistant/Gemini) are mainstream accessibility tools for motor-impaired and blind users, and are increasingly LLM-backed. **Emerging:** agentic browsers / “computer-use” agents that read the accessibility tree and act on pages could complete tasks for users who cannot navigate a bad UI — genuinely promising but not yet a mature, reliable accessibility product.

7. Cross-cutting takeaways & maturity ranking for the team

If you take five things from this report, take these.

1. **“AI for scale + human for correctness” is the pattern everywhere.** Across all seven use cases, the credible products keep a human in the loop — mandatory for sign language and visual interpreting, advisable for text simplification, and built into the leading auditing and captioning tools. Fully-automated “AI compliance” is not real yet, and claiming it has already drawn a $1M FTC penalty.
2. **The two gating technical problems are hallucination and data.** Meaning-drift/hallucination gates alt text, simplification, and visual description (a confidently wrong output is worse than none). Data scarcity gates sign language (annotated multi-view corpora are rare and expensive).
3. **Where AI adds the most genuine value over rule-based tools is auditing (Use Case 2)** — the computer-vision/semantic layer targets exactly the 40–70% of WCAG failures deterministic scanners cannot judge, while keeping humans on the subjective calls.
4. **The lowest-risk, fastest wins are alt text (1), captioning/transcription (3), and cognitive/reading support (6)** — all essentially hosted-model integration problems today.
5. **Be careful with overlays (4)** — technically trivial to deploy, but the “automated conformance” claim is legally and technically discredited. If the team builds anything overlay-like, build the honest, hybrid, human-reviewed version.

Maturity & build-readiness at a glance

Use case	Complexity	Buildable today?	Risk / caution
1. Alt text & image descriptions	Low–Med (integrate)	Yes, mature	Hallucination; missing page context
2. AI auditing & testing	High	Yes, as augmentation	Not a full-conformance replacement
3. Captioning & transcription	Med (captions)	Yes, mature	Accuracy on names/accents; audio description still emerging
4. Overlays / auto-remediation	Low to deploy	Partly — claim is the trap	Discredited; regulatory + legal risk
5. Sign-language avatars	High	Partially (constrained)	Not interpreter-quality; Deaf-community concerns
6. Cognitive / reading support	Low–Med	Yes, fully	Meaning-drift in simplification
7. Conversational / visual assistance	Med–High	Yes (assistants); partly (agents)	Hallucination in high-stakes description

Suggested starting point for a proof-of-concept: if the team wants a high-value, low-risk first build, alt-text generation (1) or AI-assisted captioning (3) will demonstrate real user benefit quickly on

> existing APIs; AI-assisted auditing (2) is the most defensible “AI beats the rule engine” story to invest in for the longer term.

8. Full list of references

All links were resolved during research (August 2026). Sources are grouped by the section they support. Items marked with a caution note in the body are repeated here for transparency.

Q1–Q3 — Foundations, legal & business case, scope

- W3C/WAI — WCAG 2 Overview (versions, POUR, conformance levels): w3.org/WAI/standards-guidelines/wcag
- W3C/WAI — What's New in WCAG 2.2: w3.org/WAI/standards-guidelines/wcag/new-in-22
- W3C/WAI — WCAG 3 Introduction: w3.org/WAI/standards-guidelines/wcag/wcag3-intro
- W3C/WAI News — WCAG 3 Working Draft (March 2026): w3.org/WAI/news/2026-03-03/wcag3
- W3C Blog — WCAG 2.1 is a Recommendation (2018): w3.org/blog/2018/wcag21-rec
- WHO — Disability Fact Sheet (1.3 billion / 16%): who.int/…/disability-and-health
- Jackson Lewis — DOJ extends Title II web-accessibility deadlines (April 2026; HHS May 2026): jacksonlewis.com/insights/…
- UsableNet — Title II compliance timeline: blog.usablenet.com/title-ii-compliance-deadline-2026
- UsableNet — ADA web lawsuit trends (5,000+ in 2025): blog.usablenet.com/ada-web-lawsuit-trends-2026
- Level Access — European Accessibility Act overview (in force June 28, 2025; EN 301 549 → WCAG 2.1 AA): levelaccess.com/compliance-overview/european-accessibility-act-eaa
- Return on Disability (via TestParty) — $13T disability market: testparty.ai/blog/disability-market-spending-power
- Kaleidoscope Group — the UK Purple Pound (~£274B): kaleidoscope.group/what-is-the-purple-pound

Q4 — Non-AI best practices

- Deque — Automated Accessibility Coverage Report (57% auto / 43% manual; 16-of-50 criteria): deque.com/automated-accessibility-coverage-report
- Deque — axe-core (deterministic rule engine): deque.com/axe/axe-core
- Deque — Top 5 Rules of ARIA ("no ARIA is better than bad ARIA"): deque.com/blog/top-5-rules-of-aria
- W3C/WAI — Evaluating Accessibility ("no tool alone can determine conformance"): w3.org/WAI/test-evaluate
- W3C — WCAG-EM Evaluation Methodology: w3.org/TR/WCAG-EM
- WebAIM — Screen Reader User Survey #10 (JAWS/NVDA/VoiceOver usage): webaim.org/projects/screenreadersurvey10
- Section508.gov — ACR / VPAT FAQ: section508.gov/sell/acr-vpat-faq
- Deque — EN 301 549 overview: deque.com/en-301-549-compliance
- W3C/WAI — ARIA Authoring Practices Guide (APG): w3.org/WAI/ARIA/apg
- eslint-plugin-jsx-a11y (static a11y linting): github.com/jsx-eslint/eslint-plugin-jsx-a11y

Use Case 1 — Alt text / image descriptions

- Be My Eyes — Introducing Be My AI: bemyeyes.com/blog/introducing-be-my-ai
- Microsoft Source — Azure AI image captioning: news.microsoft.com/source/features/ai/azure-image-captioning
- Microsoft Learn — Azure AI Vision (what's new): learn.microsoft.com/…/computer-vision/whats-new
- Google Blog — Get image descriptions: blog.google/…/get-image-descriptions
- Google Accessibility Help — image descriptions in Chrome: support.google.com/accessibility/answer/9311597

Use Case 2 — Automated accessibility testing

- Deque — axe AI: deque.com/axe/ai
- Deque — Machine learning in accessibility testing: deque.com/blog/deque-brings-machine-learning-to-accessibility-testing
- Evinced — Technology: evinced.com/technology
- Siteimprove — Accessibility Conformance Report: siteimprove.com/acr

Use Case 3 — Captioning / transcription / audio description

- 3Play Media — AI-enabled solutions (2026 ADA): 3playmedia.com/news/launches-ai-enabled-solutions
- Verbit — AI transcription & captioning: verbit.ai
- Google Research — On-device Live Caption: research.google/blog/on-device-captioning-with-live-caption
- Microsoft Learn — Azure AI Speech: learn.microsoft.com/…/speech-service/overview

Use Case 4 — Overlays / automated remediation

- Overlay Fact Sheet (expert criticism, 1,000+ signatories): overlayfactsheet.com/en
- FTC — Order requiring accessiBe to pay $1M (Jan 2025): ftc.gov/…/2025/01/ftc-order-requires-online-marketer-pay-1-million…
- FTC — Final order approved (April 2025): ftc.gov/…/2025/04/ftc-approves-final-order-requiring-accessibe-pay-1-million
- AudioEye — How the technology works: audioeye.com/post/how-audioeyes-accessibility-technology-works
- UserWay — Web accessibility solutions: userway.org
- Law Office of Lainey Feingold — analysis of the FTC/accessiBe action: lflegal.com/2025/01/ftc-accessibe-million-dollar-fine

Use Case 5 — Sign-language translation & avatars

- Signapse — home: signapse.ai
- Signapse — How our AI works: signapse.ai/post/how-does-our-ai-technology-work
- Hand Talk — App: handtalk.me/en/app (ML architecture proprietary/undocumented)
- SignAll — Technology: signall.us/technology
- World Federation of the Deaf & WASLI — Statement on signing avatars: wfdeaf.org/wfd-wasli-issue-statement-signing-avatars
- TechCrunch — SLAIT pivots to AI lessons (2023): techcrunch.com/2023/05/11/slait-pivots…

Use Case 6 — Cognitive / reading support

- Microsoft — Immersive Reader: support.microsoft.com/…/use-immersive-reader-in-microsoft-edge
- Everway (Texthelp) — Read&Write for Education: everway.com/products/read-and-write-education
- Recite Me — Assistive Toolbar: reciteme.com/product/assistive-toolbar
- W3C — WCAG 3.0 Working Draft: w3.org/TR/wcag-3.0

Use Case 7 — Conversational / voice / visual assistance

- Be My Eyes — home: bemyeyes.com
- Aira — home: aira.io
- Aira — AI Visual Interpreter (Project Astra / Gemini): aira.io/ai

- Voiceflow — conversational AI platform: voiceflow.com

Verification notes. WCAG version dates, the WHO 1.3-billion figure, the EAA in-force date (June 28, 2025), the DOJ Title II rule and its April 2026 one-year deadline extension, and the accessiBe FTC settlement were all checked against primary or reputable secondary sources. Figures for disability spending power ($13T) and the Purple Pound (£274B) are advocacy/industry estimates (directional, not audited). The “Bronze/Silver/Gold” WCAG 3 tier names, widely repeated online, are *not* confirmed on the current official W3C page and should be treated as not finalized. Hand Talk’s underlying ML architecture is proprietary and not publicly documented. This report is a research brief, not legal advice; accessibility obligations vary by jurisdiction and should be confirmed with qualified counsel for any specific product.